clature be retained until a wider survey of vertebrate and invertebrate pigments has been accomplished. It may then appear that further designations are useful, based upon whatever general distinctions have emerged. At present, however, the attachment of specific designations to each invertebrate pigment to be extracted ('cephalopsin', 'euphausiopsin') can only lead to confusion. It is easier to make new nomenclature than to get rid of old. The less nomenclature we make now, the less we shall have to regret later.

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DIRECTIONS OF MAGNETIZATION IN THE CARBONIFEROUS GLACIAL VARVES OF AUSTRALIA

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THE varved sediments in the Upper Kuttung Series of New South Wales have some interesting magnetic characteristics. Their average magnetic inclination is 84°, indicating that they were deposited in a high latitude and is therefore in agreement with their glacial origin. The pole positions consistent with their average direction of magnetization are near Lord Howe Island and Madeira, and are a very long distance away from equivalent determinations from Britain and the United States.

Varves were laid down in lakes in the vicinity of glaciers under conditions favourable to their becoming magnetized parallel to the geomagnetic field at the time of deposition. Their year-by-year banding allows samples to be taken in a known time sequence so that the secular variation of the field can be studied. Extensive work on Pleistocene and Recent varves[1] has provided information about the average direction and variation of the geomagnetic field during a small portion of these periods. Australia is fortunate in having glacial varves of very much greater age, notably in the Proterozoic and Carboniferous. Work has begun with an investigation of the stability and a determination of the average direction of the magnetization of varves in the Kuttung Series in the Hunter Valley.

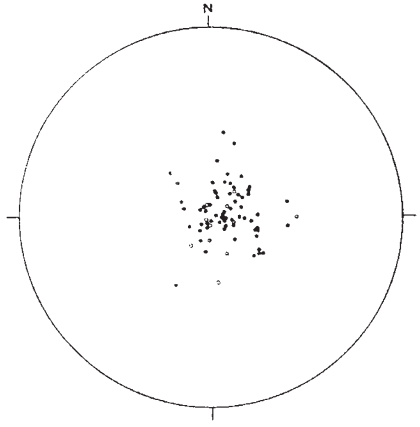


Fig. 1. Directions of magnetization in varve specimens. The directions are plotted on the lower hemisphere of a stereographic polar projection. This projection is also used in the other figures. Dots signify north-seeking poles (sites *A*, *B* and *D*) and circles south-seeking poles (site *C*). Correction is made for geological tilt, so that the plane of the projection is the bedding plane, which is assumed to have been horizontal at the time of deposition

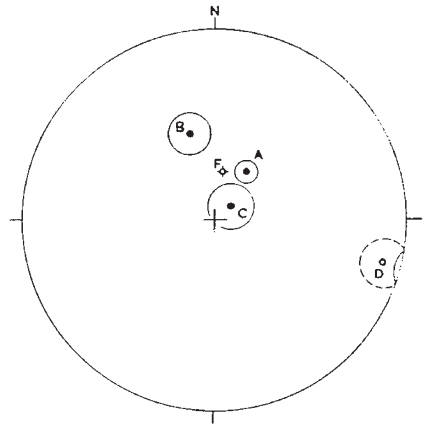


Fig. 2. Mean directions at four localities uncorrected for tilt of the beds.
The plane of the projection is the present horizontal, so that the directions are as they are in the field. Plotting is on both upper (circles) and lower (dots) hemispheres. *F* is the direction of the present Earth's field

The Upper Kuttung Series is of Upper Carboniferous age and consists of volcanics, glacial and peri-glacial deposits[2]. Results from the volcanics are being reported elsewhere[3]. The varves are for the most part grey or buff in colour, their intensities of magnetization are low (3–10 $\times$ 10^{-6} e.m.u., c.g.s. per c.c.) and their directions unstable; the varves at the type locality of Seaham exhibit such characteristics. But occasionally thin bands of pink or red colour occur,

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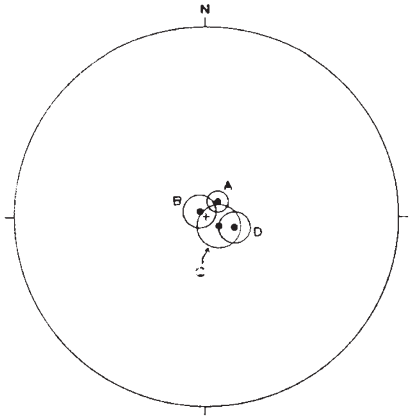


Fig. 3. Mean directions at four localities after correction for tilt. The plane of the projection is the bedding plane and plotting is on the lower hemisphere

and these have been sampled at four localities (designated *A*, *B*, *C* and *D*) in the lower Hunter Valley, seventy-five samples being taken in all. The intensities of magnetization are higher (10–50 × 10^{-6}) and the directions, which are more or less perpendicular to the bedding (Fig. 1), are stable. This has been shown by using Graham's fold test[4] as follows: the directions at the four sites are scattered when the beds are *in situ* (Fig. 2) but are grouped together after correction for geological tilt is made (Fig. 3). Thus, these directions of magnetization were imposed prior to the folding which occurred in Permian times, and may be identified with the directions of the Earth's magnetic field in this region during the Upper Carboniferous. At site *C* the north-seeking directions of magnetization are directed towards the upper bedding surface, whereas at the other three sites they point downwards in the sequence. According to current terminology these are designated normal and reversed, respectively.

It may be supposed that, when the secular variation is averaged out, the geomagnetic field approximates to a geocentric dipole. Assuming this average has been achieved, the poles of this dipole axis may be calculated and are given in Table 1 and plotted in Fig. 4. These poles are between 80° and 90° distant

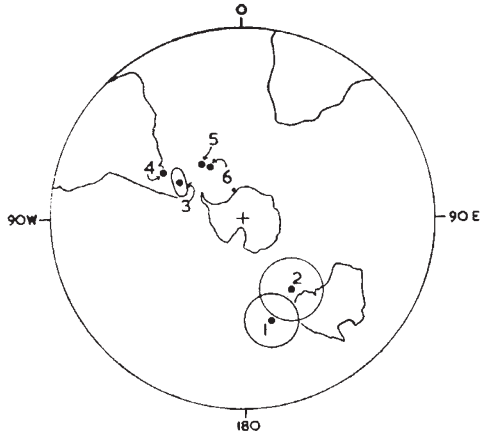


Fig. 4. Pole positions in Carboniferous times
The poles are numbered as follows: 1 and 2, Kuttung varves and lavas of Australia; 3 and 4, Naco sandstones (Pennsylvanian) and Barnett shales (Mississippian) of the United States; 5 and 6, Lower Carboniferous sediments and lavas of Britain

Table 1. Mean direction of magnetization is specified by its declination *D* measured east of geographical north, and inclination *I* positive downwards. α is the Fisher (ref. 8) angle of confidence at $P = 0{\cdot}95$. Because of reversals, the sign of this direction is uncertain. The pole errors δp and δm in the direction of, and perpendicular to, the co-latitude are at $P = 0{\cdot}95$

Mean direction of magnetization			Sampling area		Pole positions			
D	*I*	α	Lat.	Long.	Lat.	Long.	δp	δm
90°	+ 84°	6	33° S.	151° E.	165° E.	32° S.	12	12
270°	− 84°				15° W.	32° N.		

from determinations from the Mississippian[5] and Pennsylvanian[6] of the United States, and from the Lower Carboniferous of Britain[7]. Although comparatively small statistical differences may separate the Carboniferous poles of Britain and the United States, it seems very probable that for these regions at this time the pole lay in what is nowadays the South Atlantic with the anti-pole in eastern Asia, and the geomagnetic field over these regions was substantially dipolar. The complete disagreement of the Kuttung poles suggests either that Australia has moved relative to Britain and the United States, or that the geomagnetic field, although approximately dipolar over a large part of the northern hemisphere, was not so in the southern hemisphere.

The preceding argument does not assume any correlation of the magnetic and rotational axes of the Earth, but in order to make comparisons between the palæoclimatic and palæomagnetic data, let it be supposed that when averaged over several thousands of years the geocentric dipole has always been directed along the axis of spin. This may be referred to as the geocentric axial dipole hypothesis. The geographical latitudes of eastern Australia and western Europe now become approximately 80° and 70°, respectively. Since the pioneer work of Sir Edgeworth David, the climatic conditions under which the Upper Kuttung Series were laid down have always been regarded as cold or glacial—conditions which are most likely to have been attained in high latitudes. Western Europe, on the other hand, provides no evidence of glaciation, and geological opinion favours warm and wet conditions at this time. The agreement between the palæoclimatic and palæomagnetic evidence at the same place is consistent with the geocentric axial dipole hypothesis. On the other hand, the contrast between these two lines of evidence when antipodal regions are considered suggests that either western Europe and eastern Australia have moved relative to each other since the Carboniferous, or that this hypothesis is false.

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